

TWINKIT · DESIGN DECISION RECORD

TwinKit recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	TWK-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/twinkit		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted

■ Context

TWK-DDR-001 and `docs/REVIEW.md` (sessions of 2026-09-25) listed items that were adopted for TRL 3 pending Amish's review, or proposed and awaiting Amish, each with a recommendation. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item that carried a recommendation is therefore decided as recommended. Where a recommendation named one of several options, that option is the decision. Items with no recommendation stay open. TRL 4 remains on hold by Amish's instruction, and `trl` and `trl_target` stay at 3.

■ Options considered

The options for each item are those listed in `docs/REVIEW.md` (TRL 2 and TRL 3 sessions) and in TWK-DDR-001.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Radio	8-channel LoRaWAN concentrator (SX1302 or SX1303 class)	Status wording only in TWK-PRC-001, TWK-PRB-001, TWK-DDR-001 and bom/bom.csv ; no design change
D2	Computer	4 GB Raspberry Pi 5 class board	Status wording only
D3	Database	TimescaleDB	Status wording only
D4	Twin platform	Lightweight open-source stack plus a TwinKit twin service	Status wording only
D5	First example twin	FieldNode battery and solar charge	Status wording only; cross-repo action for FieldNode (adaptive data rate, below)
D6	Twin file schema	Project, model, and per channel an id, node, part (BOM number), unit, expected value, tolerance and hold time	Status wording only
D7	Backup pack	Off-the-shelf 12.8 V LiFePO4 pack with built-in BMS and fuse	Status wording only
O1	Budget	Option (a): raise the budget to \$300 to keep LoRaWAN compatibility	project.yaml budget_usd 200 to 300; R14 target \$200 to \$300 (TWK-REQ-001 v0.4); R14 not met to met (\$290); README budget line; TWK-PRB-001 constraint; bom/bom-notes.md ; concept blueprint key figure
N1	R1 at SF9	Option (a): keep the 1 % target and require adaptive data rate so near nodes use SF7 or SF8	R1 restated in TWK-REQ-001 v0.4; adaptive data rate added as a design rule in TWK-PRC-001 v0.4; TWK-CAL-001 v0.2 judges R1 on the SF7 to SF9 mix: 1.02 % (not met) to 0.21 % (met) [A6]; blueprint key figure updated
N2	R12 load profile	Define R12 at the normal light load and schedule heavy database jobs for cool hours; a larger or metal enclosure is the fallback if tests show throttling	R12 restated in TWK-REQ-001 v0.4; cool-hours scheduling added as a design rule in TWK-PRC-001 v0.4; TWK-CAL-001 v0.2 [F6]: at risk to met (69.1 °C, 15.9 K margin at light load). The enclosure fallback depends on TRL 4 tests and is on hold

No geometry changed, so cad/src/model.py , bom/bom.csv prices and drawing TWK-DWG-001 (Rev P1) are unchanged in content; the model, drawing and media were regenerated only for the site address change. No reworded pitch or problem line was recommended, so the pitch and problem text in project.yaml and README.md stay as they were.

Table 2. Items still open.

#	Item	Status
O2	Co-design partner outside the lab (a small water utility, a makerspace or a municipal team). No recommendation was made.	Proposed, awaiting Amish

■ Consequences

- Requirement status (TWK-CAL-001 v0.2, Table 2): 0 not met, 0 at risk, 8 met by calculation, 7 met by design, 1 not verifiable at TRL 3 (R13, setup time). Before: 2 not met (R1, R14), 1 at risk (R12).

- The R14 margin is small (\$10) and rests on indicative prices, not supplier quotes.
- R12 is met on assumed values for the processor's thermal resistance and throttle point; only a TRL 4 thermal test can confirm them.
- Cross-repo: FieldNode nodes must accept LoRaWAN adaptive data rate commands for R1 to hold. This is listed under "Cross-repo actions" in docs/REVIEW.md ; no other repo was edited.
- TRL 4 work implied by these decisions (a timed setup trial, thermal and packet-loss tests, supplier quotes, a trial of the enclosure fallback) is decided in principle but on hold, because TRL 4 is on hold by Amish's instruction.